

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
df = pd.read_csv("/content/Titanic-Dataset.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

Next steps:

Generate code with df

New interactive sheet

```
df.shape
```

(891, 12)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null     int64
1   Survived      891 non-null     int64
2   Pclass        891 non-null     int64
3   Name          891 non-null     object
4   Sex           891 non-null     object
5   Age           714 non-null     float64
6   SibSp         891 non-null     int64
7   Parch         891 non-null     int64
8   Ticket        891 non-null     object
9   Fare          891 non-null     float64
10  Cabin         204 non-null     object
11  Embarked      889 non-null     object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

df.describe()

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

```
df.isnull().sum()
```

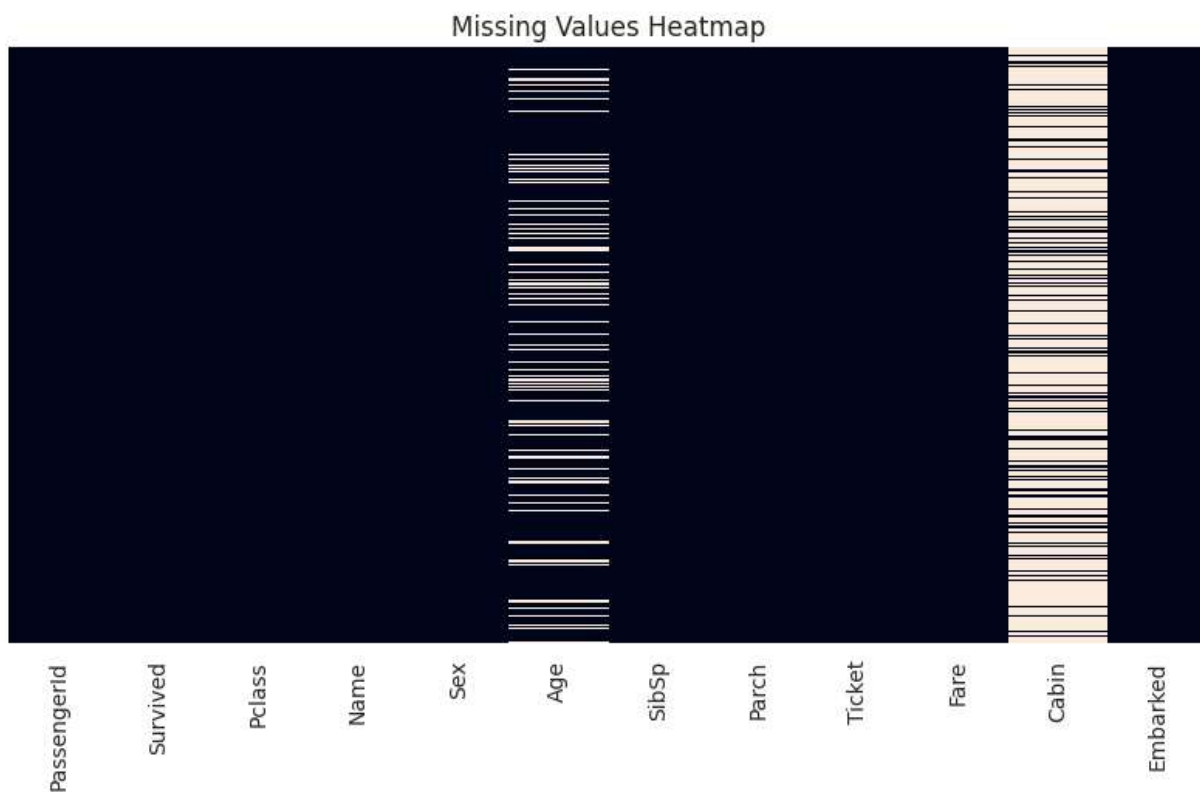
	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

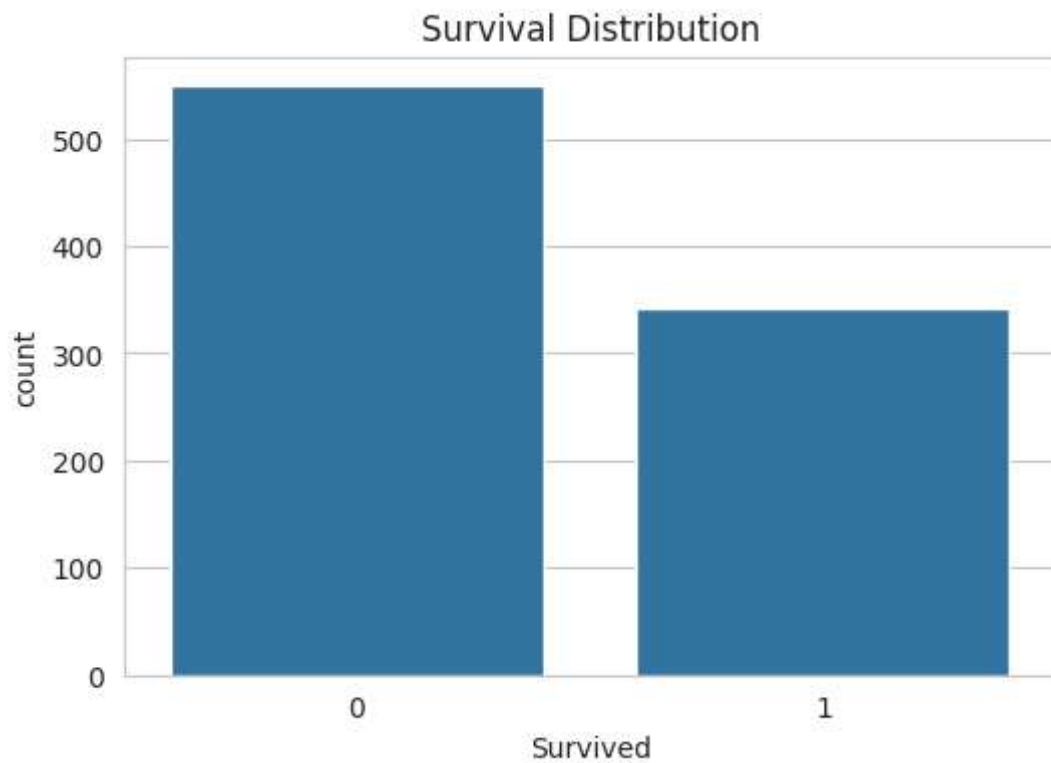
```
print(df["Survived"].value_counts())
print(df["Sex"].value_counts())
print(df["Pclass"].value_counts())
```

```
Survived
0    549
1    342
Name: count, dtype: int64
Sex
male    577
female  314
Name: count, dtype: int64
Pclass
3     491
1     216
2     184
Name: count, dtype: int64
```

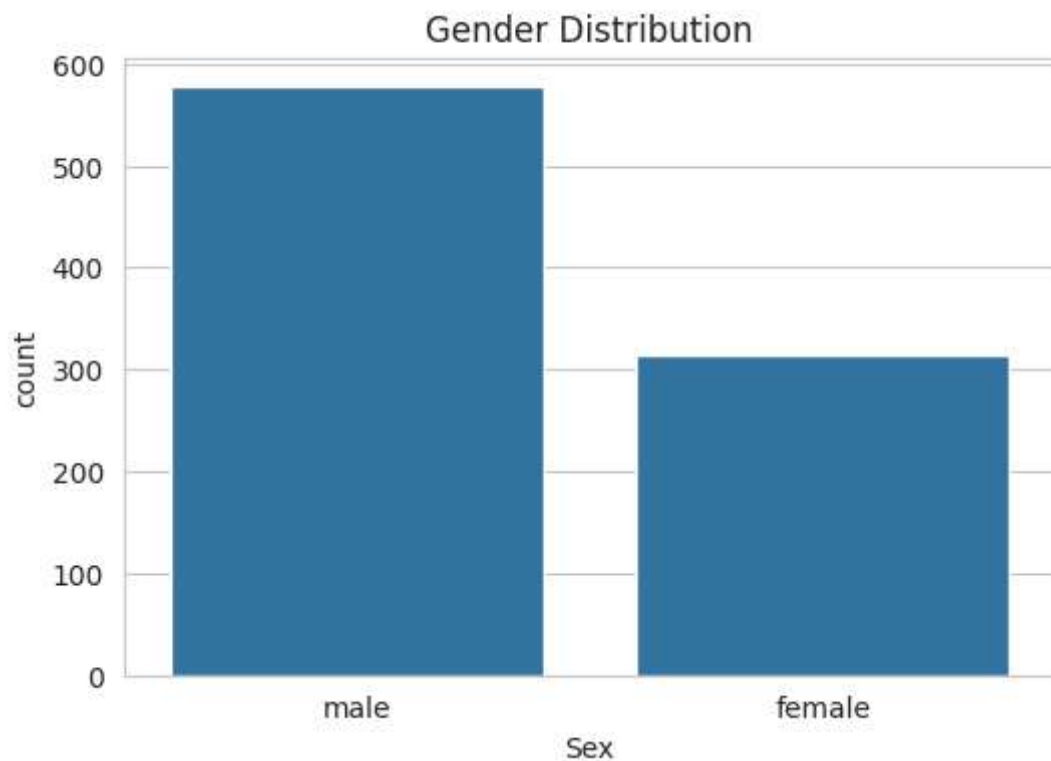
```
plt.figure(figsize=(10,5))
sns.heatmap(df.isnull(), cbar=False, yticklabels=False)
plt.title("Missing Values Heatmap")
plt.show()
```



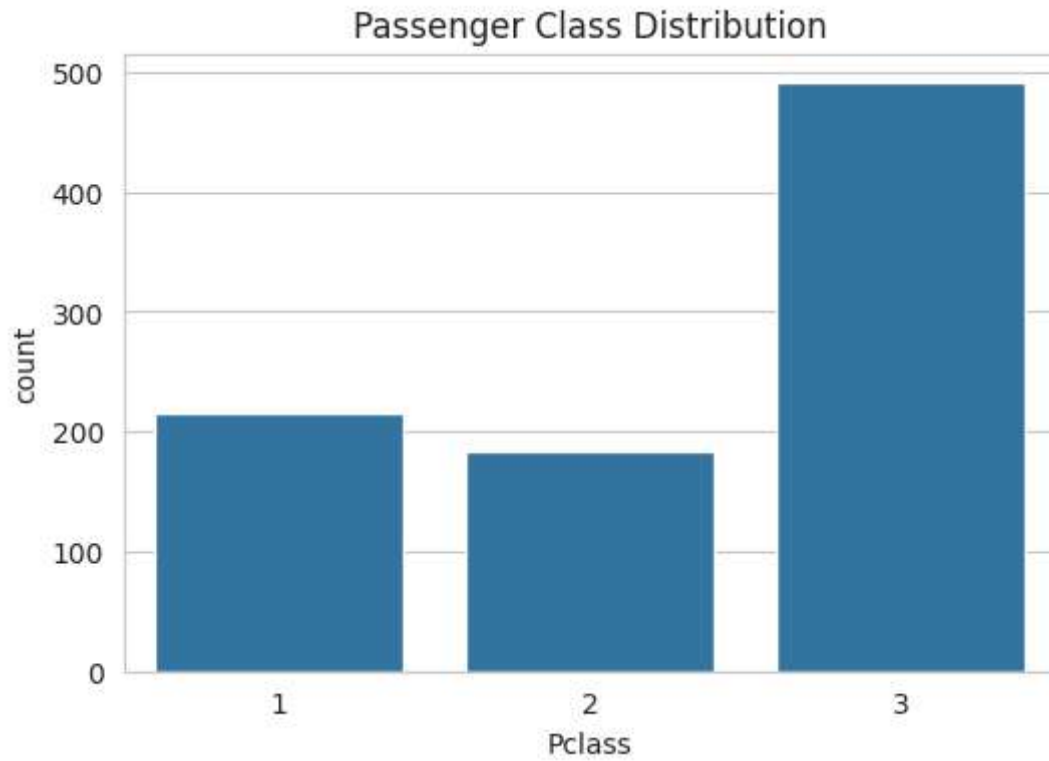
```
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution')
plt.show()
```



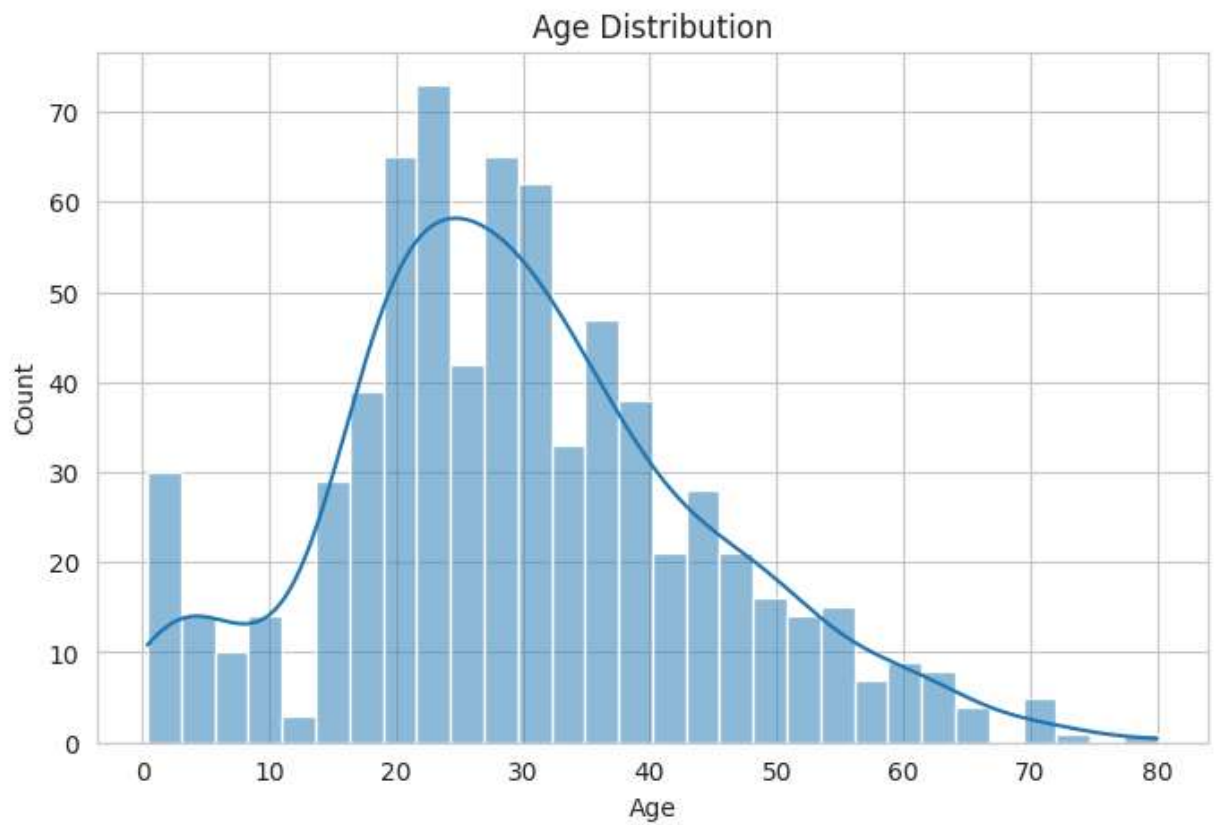
```
plt.figure(figsize=(6,4))  
sns.countplot(x='Sex', data=df)  
plt.title('Gender Distribution')  
plt.show()
```



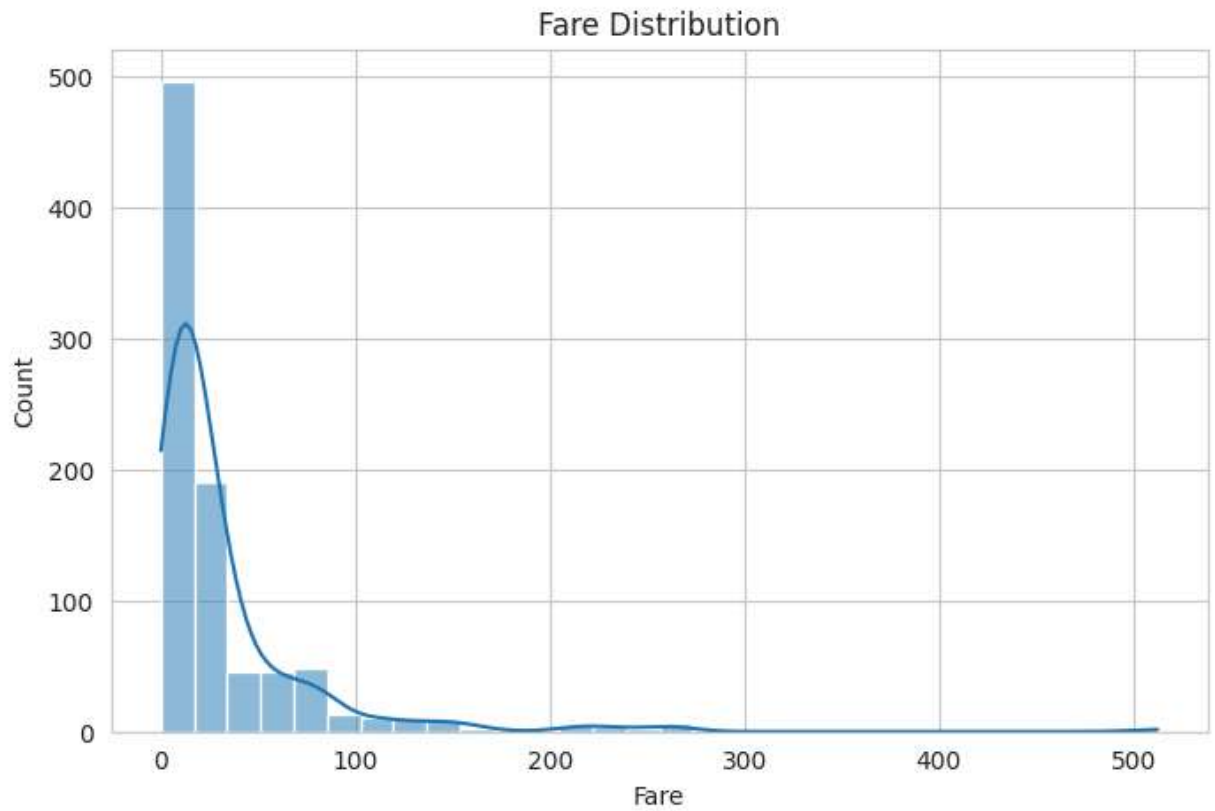
```
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', data=df)
plt.title('Passenger Class Distribution')
plt.show()
```



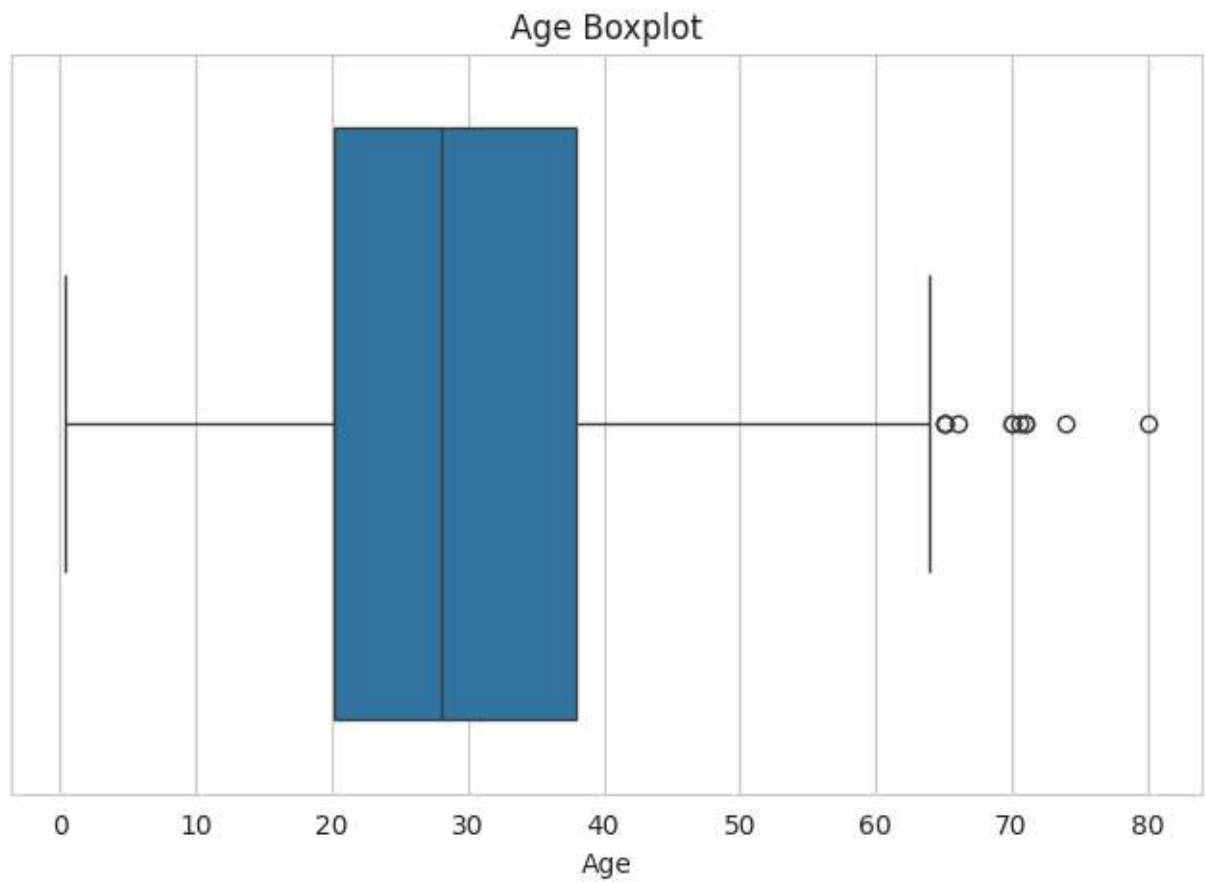
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title('Age Distribution')
plt.show()
```



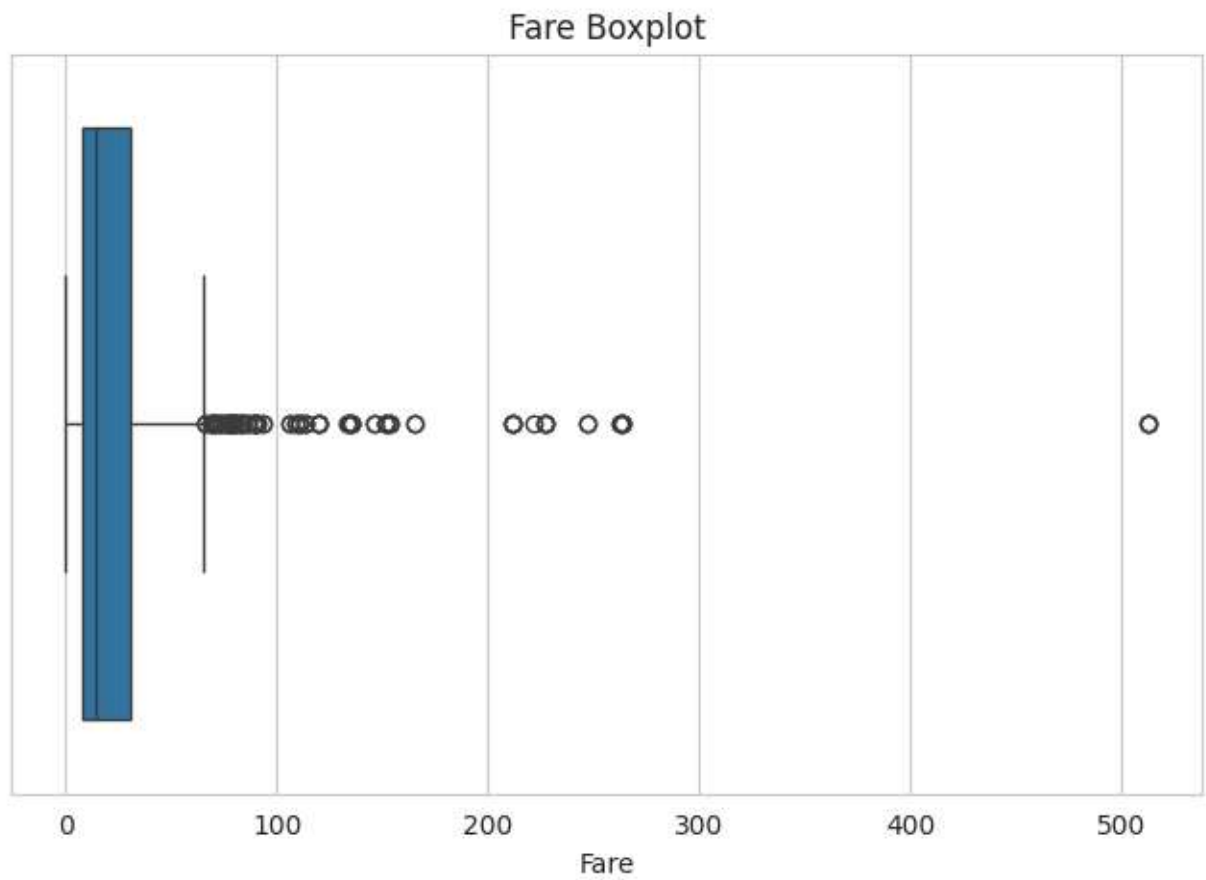
```
plt.figure(figsize=(8,5))
sns.histplot(df['Fare'], bins=30, kde=True)
plt.title('Fare Distribution')
plt.show()
```



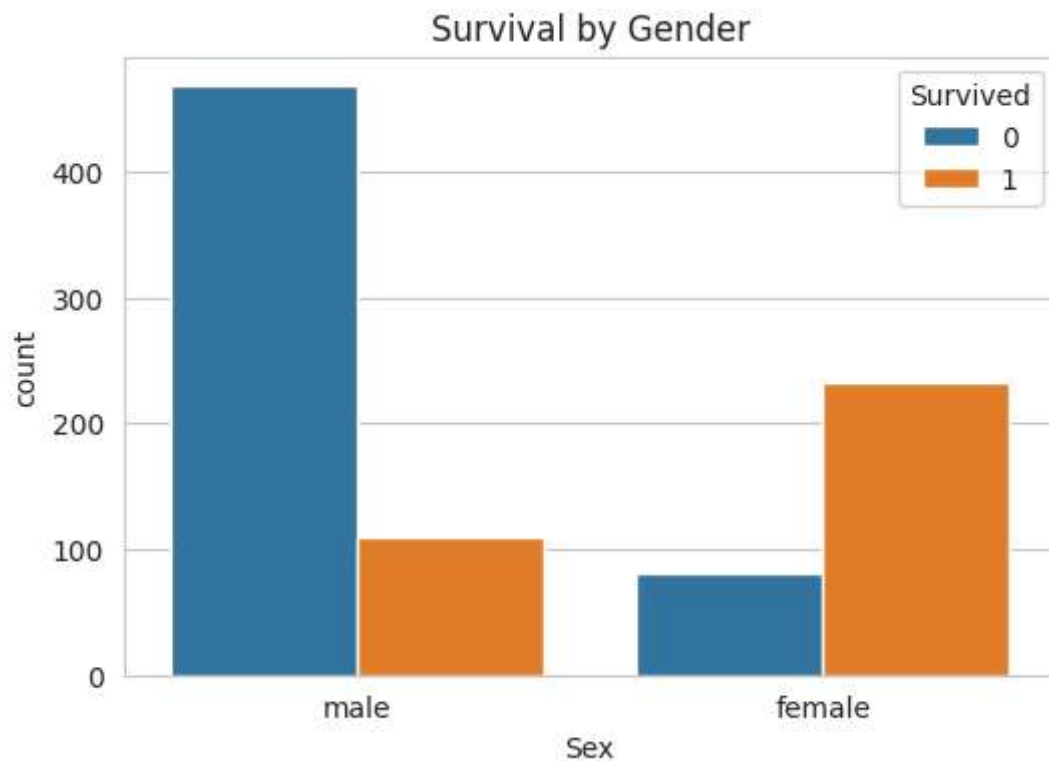
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Age'])
plt.title('Age Boxplot')
plt.show()
```

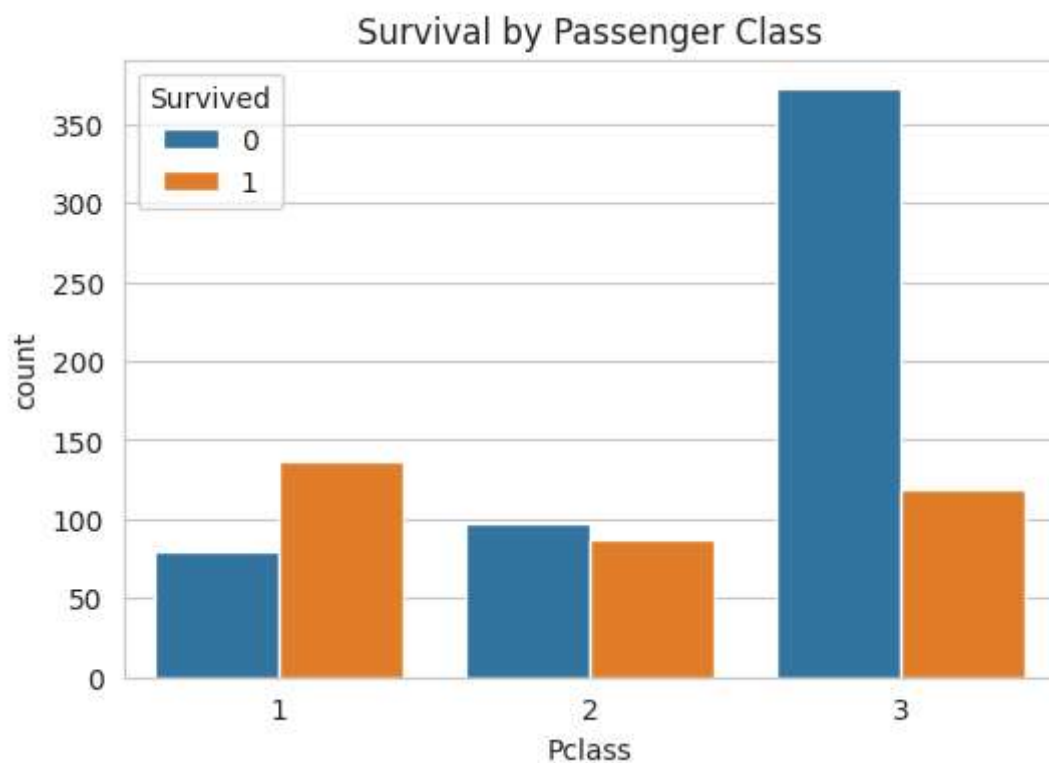
```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Fare'])
plt.title('Fare Boxplot')
plt.show()
```



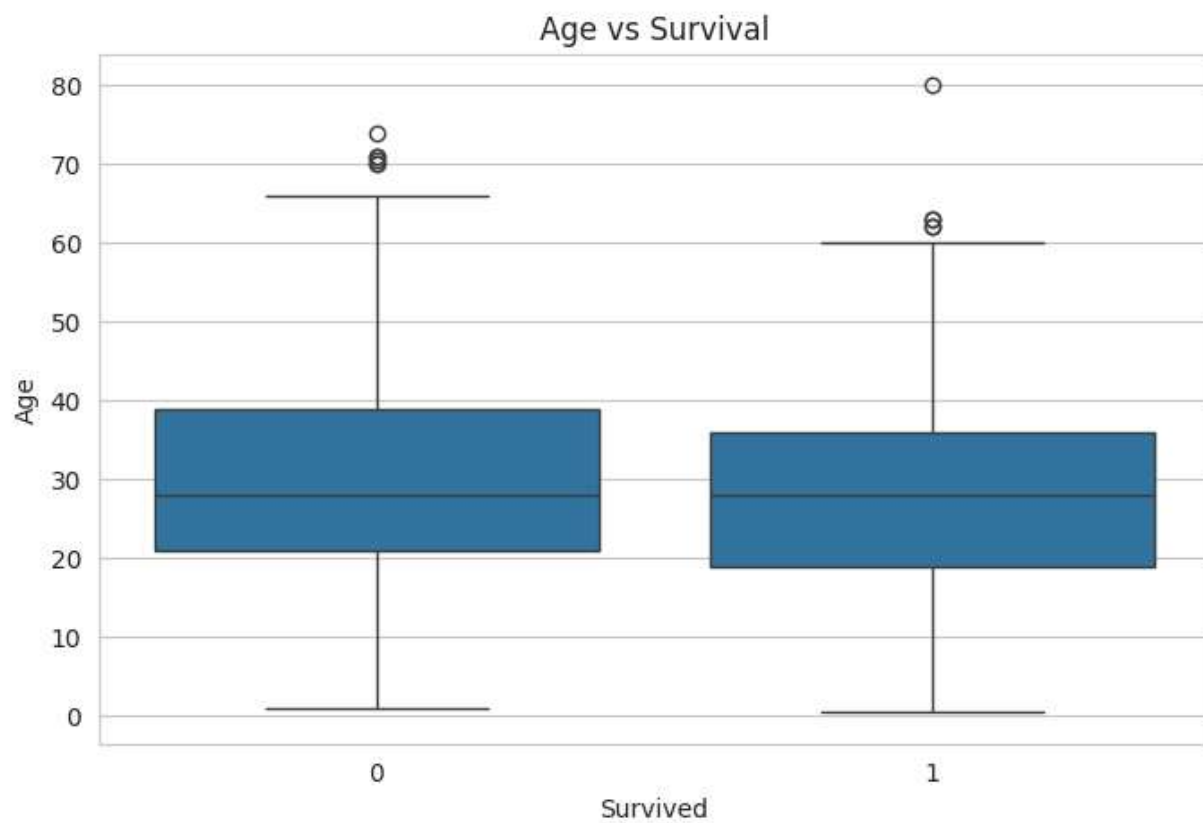
```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.show()
```



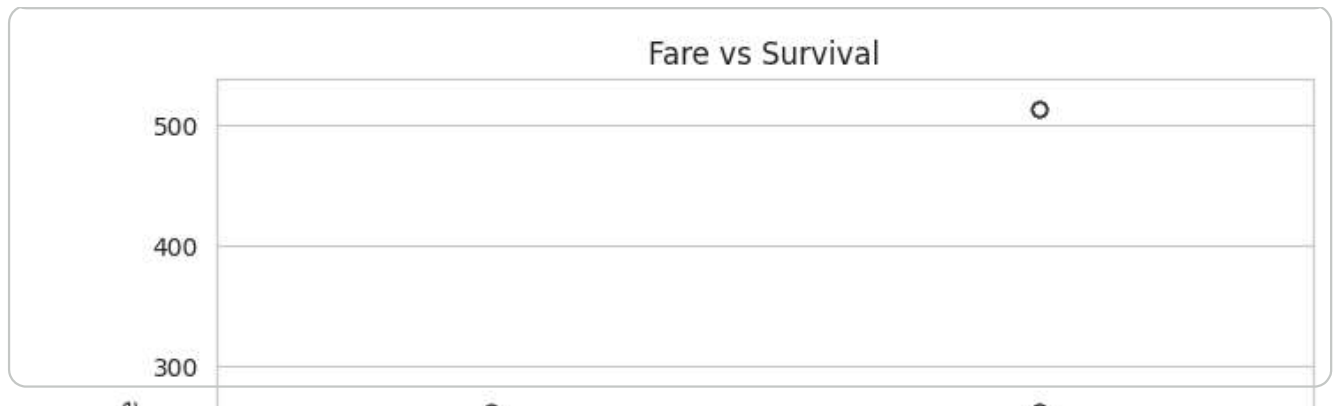
```
plt.figure(figsize=(6,4))  
sns.countplot(x='Pclass', hue='Survived', data=df)  
plt.title('Survival by Passenger Class')  
plt.show()
```



```
plt.figure(figsize=(8,5))
sns.boxplot(x='Survived', y='Age', data=df)
plt.title('Age vs Survival')
plt.show()
```



```
plt.figure(figsize=(8,5))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare vs Survival')
plt.show()
```



```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title('Age vs Fare')
plt.show()
```

